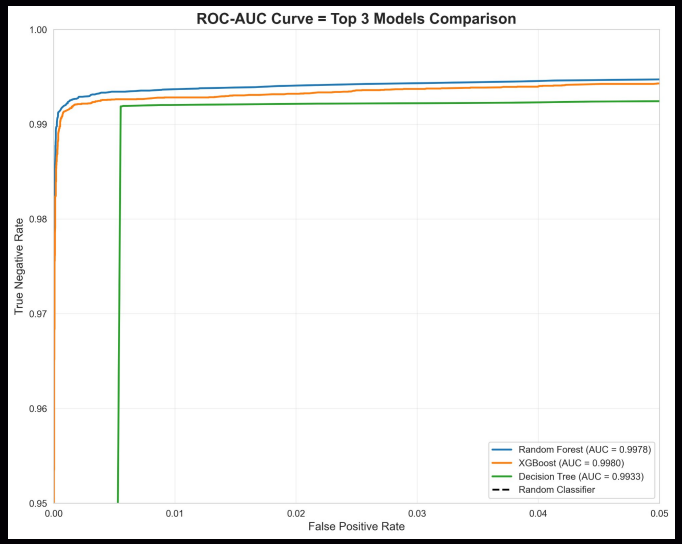
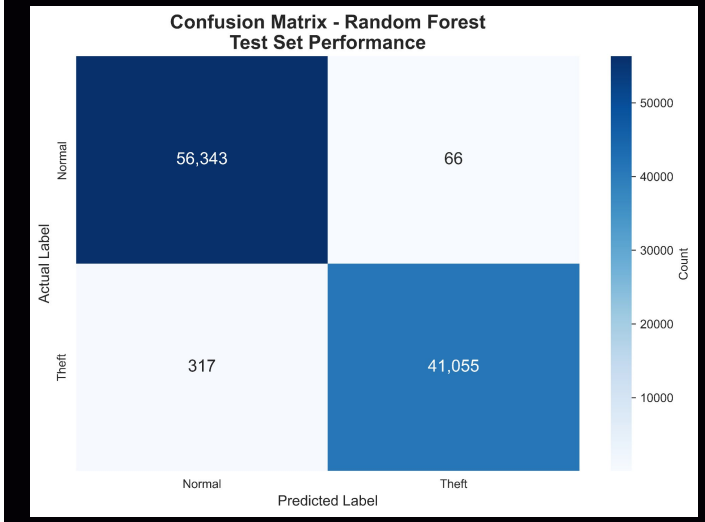
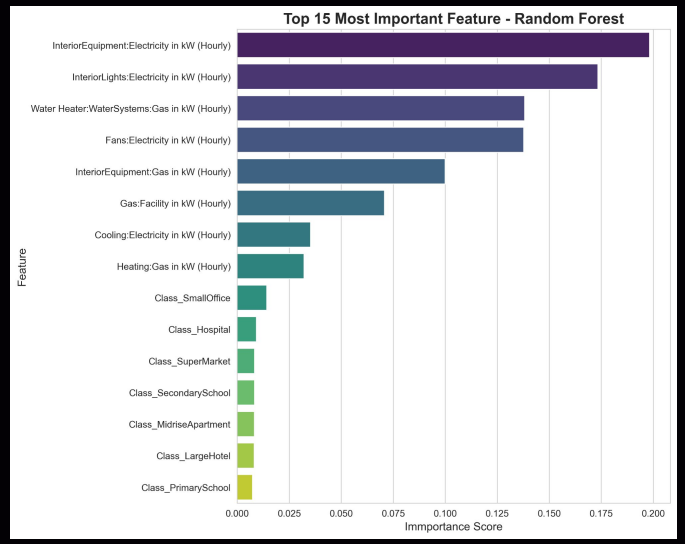
MODEL PERFORMANCE METRICS



ROC / AUC CURVE



CONFUSION MATRIX



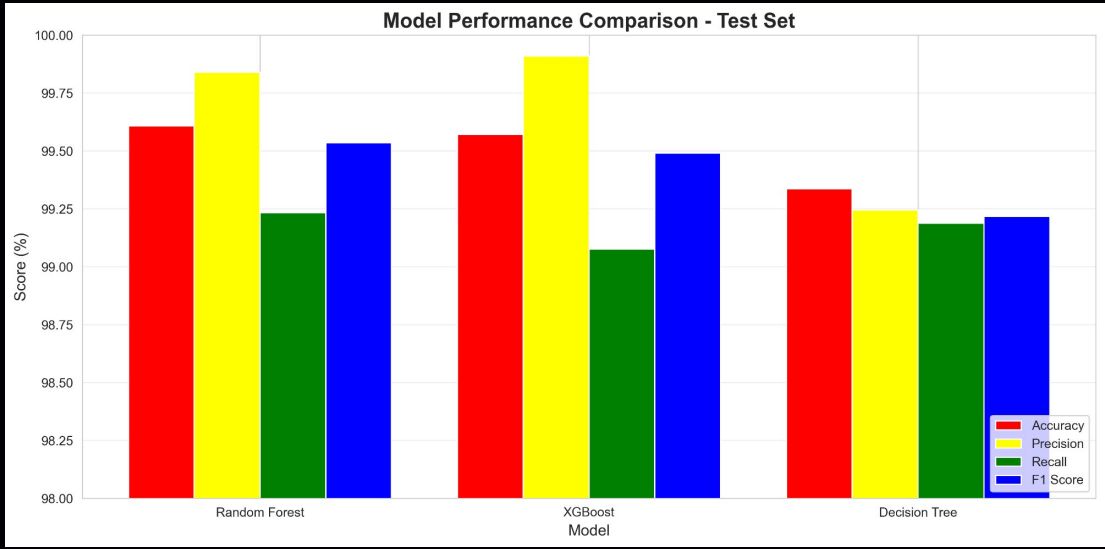
FEATURE IMPORTANCE

MODEL COMPARISON

Model Performance Comparison - Test Set

Model	F1-Score	Precision	Recall	ROC-AUC
Random Forest	99.54%	99.84%	99.23%	99.78%
XGBoost	99.49%	99.91%	99.08%	99.80%
Decision Tree	99.22%	99.25%	99.19%	99.33%

MODEL PERFORMANCE COMPARISON TABLE



MODEL PERFORMANCE COMPARISON — TEST SET (BAR CHART)

CLASSIFICATION REPORTS

Random Forest - Classification Report (Test Set)

	Precision (%)	Recall (%)	F1-Score (%)	Support
Normal	99.44	99.88	99.66	56409.0
Theft	99.84	99.23	99.54	41372.0

RANDOM FOREST — CLASSIFICATION REPORT

Random Forest - Classification Report (Test Set)

	Precision (%)	Recall (%)	F1-Score (%)	Support
Normal	99.44	99.88	99.66	56409.0
Theft	99.84	99.23	99.54	41372.0

TOP 3 MODELS — CLASSIFICATION REPORT